Disaster Exposure as an Instrument:

Evaluating Income Effects on Education in Nepal

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Abstract

This paper explores the use of disaster exposure as an instrument to study the relationship between household income and educational investment in Nepal. Using predicted ground motion acceleration from 58 earthquakes between 2016 and 2018, I measure seismic exposure for 7,918 households observed in a two-year panel survey. Controlling for household characteristics and fixed effects, I find both first- and second-stage results significant. I suspect endogeneity and data availability as major concerns. Precise household-level exposure, and data on supply-side constraints such as damage to schooling infrastructure could improve future studies.

1 Question

Does a family's income explain how much they invest in their children's education? And can the random nature of earthquake events help identify this relationship?

Relationship between income and educational spending is far more nuanced. Decisions about investing in a child's education are shaped not only by financial constraints but also by parental

preferences, expectations about future returns, cultural norms, and other difficult-to-observe factors. When a household spends little on education, it is therefore unclear whether this reflects an inability to pay or a belief that such investments may not be worthwhile.

Since these represent fundamentally different situations, each requiring distinct policy responses, understanding the mechanisms behind educational investment is critical. This motivates my present study: by leveraging household data and variation in seismic exposure from recent earthquakes in Nepal, I examine whether disaster exposure can serve as a valid instrumental variable to identify the causal effect of income on educational investment.

2 Motivation

Two key motivations drive my interest in this topic.

First, a child born into a poor household may face lifelong disadvantages. If families lack the means to invest in education, their children might perform worse in the labor market and earn less in the future. This disadvantage can compound across generations, creating cycles of poverty that are difficult to break. I believe that understanding the true relationship between income and educational investment is critical for uncovering the roots of social inequality—and, in the long run, for informing efforts to build more equitable societies.

Second, disasters offer a stark illustration of how poverty can have lasting effects, possibly through accumulation of human capital. When a society is struck by a natural shock, such as an earthquake, wealthier families are often better able to shield themselves. Disasters can deepen income disparities, and these in turn may influence how much families are able—or willing—to invest

in their children's education. Studying this link can help us better design policies that prevent intergenerational losses and protect vulnerable populations during such events.

My specific interest lies in studying this relationship in Nepal, a country highly prone to earthquakes. This presents a unique opportunity to exploit the random nature of seismic shocks to study household behavior. Since earthquakes strike unpredictably, it's reasonable to assume that unobservable household traits like motivation or future expectations are not systematically related at the time of exposure. Families cannot choose to live closer to or farther from future epicenters, and urban and rural areas alike face similar risks.

What earthquakes do affect, however, is income—through loss of property, health, livelihoods, and psychological well-being. I use the income variation induced by earthquakes to isolate the causal impact of income on educational investment. This natural experiment provides a rare opportunity to examine a fundamental question. However, there remain important identification concerns, particularly regarding potential violations of the exclusion restriction and unobserved household-level shocks, which I discuss in detail in the Methodology section.

3 Literature

Previous research consistently shows household income as an important determinant of educational investment in children. Economists have long argued that families facing financial constraints tend to prioritize immediate necessities, thus under-investing in education despite its long-term benefits (Becker, 1991; Schultz, 2004). Empirical studies confirm that increases in household income generally lead to higher educational expenditures. For example, Paxson and Schady (2007) use

household expenditure surveys to demonstrate that families in Ecuador significantly increased spending on children's education following positive income shocks. Similar findings arise from rural Mexico, where conditional cash transfers raised household income, subsequently leading to improvements in children's school enrollment and expenditures on educational materials (Attanasio and Kaufmann, 2012). These expenditure-based studies underscore the direct relationship between financial resources and educational investment.

However, isolating the causal impact of income on education spending remains challenging due to confounding factors like parental motivation and future expectations. Recent studies leverage natural experiments, notably earthquakes, to provide identification. Earthquakes serve as sudden, unpredictable shocks that can severely disrupt household finances. Andrabi et al. (2020) examine the 2005 Kashmir earthquake in Pakistan, finding negative impacts on household income and children's schooling outcomes, particularly in severely affected regions. Similarly, Caruso and Miller (2015) explore Peru's 1970 Ancash earthquake, demonstrating persistent educational deficits among children exposed to the quake in utero, particularly affecting disadvantaged families and exacerbating intergenerational inequality. Paudel and Ryu (2018) find analogous long-term educational setbacks among Nepali children from low-caste households following the 1988 earthquake, reinforcing the idea that natural disasters deepen existing socio-economic divides.

Methodologically, these studies adopt mix of instrumental variable (IV) techniques and difference-in-differences (DiD) frameworks to exploit the exogenous variation provided by seismic shocks. For instance, distance from the earthquake's epicenter or intensity of ground shaking measured through peak ground acceleration have commonly served as instruments for household income (Andrabi et al., 2020; Caruso and Miller, 2015). Paudel and Ryu (2018) use DiD methods

by comparing educational outcomes before and after earthquakes across affected and unaffected regions, controlling for unobservable regional and household characteristics. These methods try to find causality by minimizing biases from omitted variables and self-selection.

My study contributes to this literature by uniquely combining detailed household expenditure data with precise earthquake intensity measures at the community level in Nepal. By calculating ground motion acceleration for each community-earthquake pair, I offer a measure of seismic exposure for IV estimates. Moreover, my paper investigates how short-term income disruptions from earthquakes impact educational spending decisions, bridging gaps in literature's understanding of disaster-driven inequality in human capital formation.

4 Data

I draw on two key data sources for this study: a two-year household panel survey covering 5,654 families, and administrative records of earthquakes that struck Nepal between June 2016 and June 2018. I briefly describe both below.

The household panel data come from the Nepal Household Risk and Vulnerability Survey (NHRVS), a nationally representative survey jointly administered by the World Bank and the Government of Nepal. The NHRVS was conducted over three consecutive years—Wave 1 in 2016, Wave 2 in 2017, and Wave 3 in 2018—with each round fielded between June and August. The survey was explicitly designed to follow the same households over time and to capture how they respond to a variety of shocks, including the 2015 Gorkha Earthquake. I only use the last two surveys for this study.

The sampling frame covers 50 districts across Nepal's ecological and geographic zones. Within

these districts, 400 primary sampling units (PSUs)—hereafter referred to as ”communities”—were selected using stratified random sampling. Each community includes 15 households, resulting in a baseline of 6,000 households in Wave 1. The panel retention rate is high: 94% of households were tracked through all three waves, yielding a final panel of 5,654 households. Households that migrated permanently were replaced, and split-off households were tracked when feasible.

The earthquake data include 58 events of magnitude 4.0 or higher recorded in Nepal from June 2016 to June 2018. For each event, the dataset reports the date, time, and geographic coordinates of the epicenter. To find distance of families from the epicenter, I estimate community-level locations using Google Maps API based on village names and then calculate the geodesic distance to the epicenter.

To better capture earthquake exposure, I estimate ground motion acceleration for each community-earthquake pair using both the magnitude and the distance to the epicenter—widely considered a more accurate measure than magnitude alone. I refer to this measure as earthquake intensity, or simply intensity for ease of exposition.

For each community and year, I aggregate intensity across all earthquakes experienced in that period. This cumulative measure serves as my proxy for earthquake exposure. Naturally, this assumes that all families within a community experience the same level of exposure—a strong but necessary simplification.

5 Methodology

I assume a linear relationship between educational investment and family's income. My goal here is to estimate the parameter α in the following equation:

$$E_{itc} = \alpha Y_{itc} + \beta X_{itc} + \eta_i + \kappa_t + \delta_c + \zeta_{itc} \tag{1}$$

Where, E_{itc} represents how much family i in community c spends on education during year t. The income variable, Y_{itc}, is measured by the family's total yearly expenditure. X_{itc} are the observable household characteristics such as the education of family head, average grade level of all the family members enrolled in school, number of enrolled members, and total household size. η_i, κ_t, and δ_c account for family, time, and community-specific errors, respectively, and ζ_{itc} is the error term.

To isolate income's causal impact, I start with a first-stage regression that leverages earthquake-induced income variation:

$$Y_{itc} = \mu_1 I_{c,t} + \gamma_1 X_{itc} + \beta_{11} D_i + \beta_{12} D_t + \beta_{13} D_c + \nu_{itc} \tag{2}$$

In this model, $I_{c,t}$ is the earthquake intensity experienced by community c in the year leading upto the survey—a useful measure of seismic shock exposure. Family (D_i), time (D_t), and community (D_c) fixed effects are included to control for family, time, and community-specific errors. The term ν_{itc} is the unexplained variation.

Next, I run following second-stage regression:

$$E_{itc} = \mu_2 I_{c,t} + \gamma_2 X_{itc} + \beta_{21} D_i + \beta_{22} D_t + \beta_{23} D_c + \epsilon_{itc} \tag{3}$$

Here, notation is consistent with the first-stage equation, and ϵ_{itc} represents unexplained variation in education expenditure.

Finally, the causal impact of income on educational investment, α, is estimated by dividing the coefficients from these two stages:

$$\alpha = \frac{\mu_2}{\mu_1} \tag{4}$$

We need to consider few issues before using earthquake exposure as an instrument in my model.

First, earthquake events may influence unobserved factors that lie in the error term of Equation 1. A key concern in my setting is the potential destruction of schooling infrastructure. Since my dataset does not include information on which school a child attends, I cannot directly observe whether an earthquake damaged that school. If earthquakes reduce spending on education through school closures or reduced access—rather than through reduced household income—the exclusion restriction would be violated. To address this, I rely on community fixed effects, under the assumption that children within a given community generally attend the same school. While this is a strong assumption, it is not implausible because community schools run by government do enroll large number of students in Nepal.

Second, I do not have family wealth as observed characteristics in the model– this will likely get

impacted by earthquake. To make the matter more difficult, family's wealth can have intangible assets like health which is not fully measured by my survey. Earthquakes do damage homes or injure family members, both of which may independently affect educational investment decisions. This would again violate the exclusion restriction of my instrument. In this proposal, I assume that the effects of earthquakes on family wealth are shared collectively and that households make joint decisions about educational spending. This allows me to control for household-specific, time-invariant unobserved heterogeneity using family fixed effects. One possible violation of this assumption could occur if an individual family member is disproportionately affected—say, by injury—while others continue their education unaffected. Even so, the fixed effects should help mitigate much of this concern.

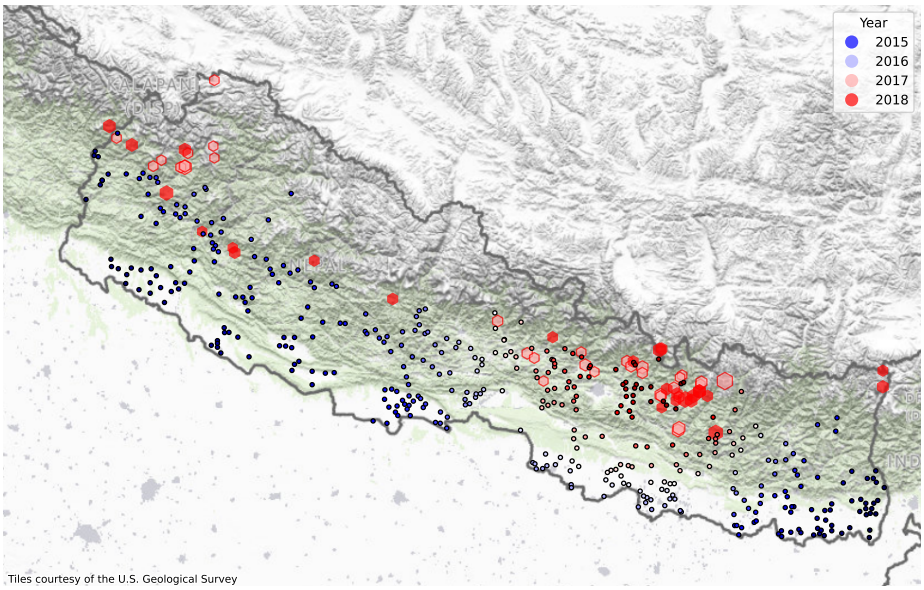
6 Results and Discussion

Table 1: Summary Statistics

	Mean	sd	Min	Max
Earthquake Intensity	94.6	11.2	78.9	120
Total Spending (NRs.)	27.5	20.1	3.51	172
Education Spending (NRs.)	2.21	3.1	0	106
Family Head Education	4.5	4.61	0	17
# Enrolled in School	1.97	1.03	1	9
Average grade enrolled	5.65	3.62	0	16
Family Size	5.46	1.97	2	22
BCN Caste (=1)	.345	.475	0	1
N	7918			

Note: This table summarizes the key variables used in the analysis. The number of observations is slightly below the full sample of 12,056 (approximately 6,000 families surveyed over two years) because families with no members enrolled in school were excluded. Nepali Rupees (NRs.) is reported in ten thousands.

Figure 1: Earthquakes and Cumulative Intensity (2017-2018)



Note: Pentagon markers represent all earthquakes of magnitude 4.0 or higher recorded by Nepal's National Earthquake Monitoring and Research Center between June 2016 and June 2018. Circle markers indicate communities from the HRVS data, with colors showing cumulative earthquake intensity—red for higher intensity, blue for lower.

Table 2 provides a balance check by comparing above- and below-median exposure households within each survey year. Across most characteristics—including household head education, number of enrolled children, and average grade level—the differences are small and statistically insignificant, suggesting that earthquake exposure is not concentrated among households with systematically different observable characteristics. The only robust imbalance appears in caste composition and, in the total spending among 2018 sample. While these differences are modest in magnitude, they indicate that earthquake shocks are not perfectly random across communities, underscoring the importance of including Family and Time fixed effects in my method.

Turning to the instrumental-variable results, Table 3 shows that earthquake intensity significantly reduces both household income and educational spending. In the first stage, a one-unit increase

Table 2: Balance Table

	2017	2018	All Sample
Total Spending (NRs.)	0.0123	-6.359***	1.311**
	(0.666)	(0.603)	(0.451)
Education Spending (NRs.)	0.252*	0.0485	0.00704
	(0.106)	(0.0911)	(0.0698)
Family Head Education	0.0381	0.333*	0.0828
	(0.148)	(0.145)	(0.104)
# Enrolled in School	0.0309	-0.0392	-0.0156
	(0.0334)	(0.0324)	(0.0232)
Average grade enrolled	-0.0397	0.0356	0.0191
	(0.117)	(0.114)	(0.0813)
Family Size	-0.0245	-0.0305	-0.133**
	(0.0638)	(0.0615)	(0.0442)
BCN Caste (=1)	0.0665***	0.119***	0.00706
	(0.0152)	(0.0150)	(0.0107)
N	3915	4003	7918

Note: This table compares mean values between families exposed to above-median and below-median earthquake intensity. Column 1 presents results for 2017, where groups are defined using intensity measured in that year; Column 2 follows the same approach for 2018. Column 3 pools data from both years and defines above/below median groups based on the combined distribution of intensity. Nepali Rupees (NRs.) is reported in ten thousands.

in intensity lowers total yearly expenditure by approximately 1.73 (in ten-thousands of Nepali rupees), indicating that seismic shocks meaningfully disrupt household economic conditions. The second stage, however, reveals that earthquake exposure also directly affects educational spending, complicating efforts to isolate the effect operating solely through income. This pattern suggests that the exclusion restriction may not hold fully in this setting, as earthquakes appear to influence education through multiple channels beyond household income.

Reasons I suspect for this are two folds: measuring earthquake exposure using distance from the epicenter may have been imprecise at the community level as household-specific variation in shock intensity could be affected by building material, injuries to family members, and other factors that correlate to wealth of families. At the same time, earthquakes likely influence educational spending through supply channels such as damage to schools which I do not have data on. Although family and community fixed effects help absorb time-invariant unobserved heterogeneity, they cannot fully address these concerns. Consequently, my estimates should be interpreted as inconclusive, underscoring the need for more granular exposure data and better information on post-disaster household conditions to isolate income effects more reliably.

Table 3: Effect of Earthquake Intensity on Total and Educational Expenditure

	Total Expense	Education Expense
	(1)	(2)
Earthquake Intensity	-1.727***	-0.060*
	(0.316)	(0.032)
Family Head Education	0.345*	-0.012
	(0.192)	(0.017)
# Enrolled in School	1.899*	0.971***
	(1.131)	(0.141)
Average Grade Enrolled	1.147**	0.175***
	(0.549)	(0.046)
Family Size	3.296***	0.025
	(0.816)	(0.064)
Community FE	✓	✓
Time FE	✓	✓
Family FE	✓	✓
R-squared	0.699	0.818
# Communities	412	412
# Observations	7,918	7,918

Note: Column 1 represents the first stage regression on Equation (2). Column 2 represents the second stage regression on Equation (3). Nepali Rupees (NRs.) is reported in ten thousands. Standard errors clustered at the community level are reported in parentheses.

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